

DBMS – PRODUCT AND CATEGORY MANAGEMENT

1. Database Design

1.1 Category Table

Attribute	Data Type	Constraint
Category_ID	INT	Primary Key
Category_Name	VARCHAR(100)	NOT NULL, UNIQUE
Description	VARCHAR(200)	—

1.2 Product Table

Attribute	Data Type	Constraint
Product_ID	INT	Primary Key
Product_Name	VARCHAR(100)	NOT NULL
Brand	VARCHAR(100)	NOT NULL
Category_ID	INT	Foreign Key
Price	NUMBER(10,2)	NOT NULL
Stock	INT	NOT NULL
Description	VARCHAR(200)	—

2. Primary and Foreign Key Relationship

Category → Product

- Category.Category_ID → Primary Key
- Product.Product_ID → Primary Key
- Product.Category_ID → Foreign Key referencing Category.Category_ID
- Relationship: One-to-Many (1:M)

Table Creation

1.Create Category Table

```
SQL> CREATE TABLE Category (  
2     Category_ID INT PRIMARY KEY,  
3     Category_Name VARCHAR(100) NOT NULL UNIQUE,  
4     Description VARCHAR(200)  
5 );
```

Table created.

2.Create Product Table

```
SQL> CREATE TABLE Product (  
2     Product_ID INT PRIMARY KEY,  
3     Product_Name VARCHAR(100) NOT NULL,  
4     Brand VARCHAR(100) NOT NULL,  
5     Category_ID INT,  
6     Price NUMBER(10,2) NOT NULL,  
7     Stock INT NOT NULL,  
8     Description VARCHAR(200),  
9     FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID)  
10 );
```

Table created.

Data Insertion

Category Records

```
SQL> INSERT INTO Category VALUES
  2 (1, 'Guitars', 'String musical instruments');

1 row created.

SQL>
SQL> INSERT INTO Category VALUES
  2 (2, 'Keyboards', 'Electronic keyboard instruments');

1 row created.

SQL>
SQL> INSERT INTO Category VALUES
  2 (3, 'Drums', 'Percussion instruments');

1 row created.

SQL>
SQL> INSERT INTO Category VALUES
  2 (4, 'Microphones', 'Audio recording microphones');

1 row created.

SQL>
SQL> INSERT INTO Category VALUES
  2 (5, 'Amplifiers', 'Sound amplification equipment');

1 row created.
```

Product Records

```
SQL> INSERT INTO Product VALUES
  2 (101, 'Yamaha Acoustic Guitar', 'Yamaha', 1, 15999.00, 20, 'Acoustic guitar for beginners');

1 row created.

SQL>
SQL> INSERT INTO Product VALUES
  2 (102, 'Fender Electric Guitar', 'Fender', 1, 45999.00, 10, 'Electric guitar');

1 row created.

SQL>
SQL> INSERT INTO Product VALUES
  2 (103, 'Casio Electronic Keyboard', 'Casio', 2, 12999.00, 15, '61-key electronic keyboard');

1 row created.

SQL>
SQL> INSERT INTO Product VALUES
  2 (104, 'Pearl Drum Set', 'Pearl', 3, 35999.00, 8, 'Complete drum set');

1 row created.

SQL>
SQL> INSERT INTO Product VALUES
  2 (105, 'Shure Vocal Microphone', 'Shure', 4, 8999.00, 25, 'Professional vocal microphone');|
```

Data Retrieval

Display Category Records

```
SQL> SELECT * FROM Category;
```

```
CATEGORY_ID
```

```
CATEGORY_NAME
```

```
DESCRIPTION
```

```
1
```

```
Guitars
```

```
String musical instruments
```

```
2
```

```
Keyboards
```

```
Electronic keyboard instruments
```

```
CATEGORY_ID
```

```
CATEGORY_NAME
```

```
DESCRIPTION
```

```
3
```

```
Drums
```

```
Percussion instruments
```

```
4
```

```
Microphones
```

```
CATEGORY_ID
```

```
CATEGORY_NAME
```

```
DESCRIPTION
```

```
Audio recording microphones
```

```
5
```

```
Amplifiers
```

```
Sound amplification equipment
```

Display Product Records

```
SQL>
SQL> SELECT * FROM Product;
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
BRAND
-----
CATEGORY_ID      PRICE      STOCK
-----
DESCRIPTION
-----
      101
Yamaha Acoustic Guitar
Yamaha

PRODUCT_ID
-----
PRODUCT_NAME
-----
BRAND
-----
CATEGORY_ID      PRICE      STOCK
-----
DESCRIPTION
-----
      1      15999      20
Acoustic guitar for beginners

PRODUCT_ID
-----
PRODUCT_NAME
-----
BRAND
-----
CATEGORY_ID      PRICE      STOCK
-----
DESCRIPTION
-----
```

```
      102
Fender Electric Guitar
Fender

PRODUCT_ID
-----
PRODUCT_NAME
-----
BRAND
-----
CATEGORY_ID      PRICE      STOCK
-----
DESCRIPTION
-----
      1      45999      10
Electric guitar

PRODUCT_ID
-----
PRODUCT_NAME
-----
BRAND
-----
CATEGORY_ID      PRICE      STOCK
-----
DESCRIPTION
-----

      103
Casio Electronic Keyboard
Casio

PRODUCT_ID
-----
PRODUCT_NAME
-----
BRAND
-----
CATEGORY_ID      PRICE      STOCK
-----
```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
BRAND
-----
CATEGORY_ID  PRICE  STOCK
-----
DESCRIPTION
-----
      3      35999      8
Complete drum set

PRODUCT_ID
-----
PRODUCT_NAME
-----
BRAND
-----
CATEGORY_ID  PRICE  STOCK
-----
DESCRIPTION
-----
     105
Shure Vocal Microphone
Shure

PRODUCT_ID
-----
PRODUCT_NAME
-----
BRAND
-----
CATEGORY_ID  PRICE  STOCK
-----
DESCRIPTION
-----
      4      8999      25
Professional vocal microphone

```

Product Update

Update 1 – Product Price

```

SQL> UPDATE Product
  2  SET Price = 16999.00
  3  WHERE Product_ID = 101;

1 row updated.

```

Update 2 – Product Stock

```

SQL> UPDATE Product
  2  SET Stock = 25
  3  WHERE Product_ID = 101;

1 row updated.

```

Product Deletion

Delete 1

```
SQL> DELETE FROM Product
      2 WHERE Product_ID = 105;

1 row deleted.
```

Delete 2

```
SQL> DELETE FROM Product
      2 WHERE Product_ID = 104;

1 row deleted.
```

Category-Wise Product Reports

```
SQL> SELECT
      2     c.Category_Name,
      3     p.Product_Name,
      4     p.Brand,
      5     p.Price,
      6     p.Stock
      7 FROM Category c
      8 JOIN Product p
      9 ON c.Category_ID = p.Category_ID
     10 ORDER BY c.Category_Name;
```

```
CATEGORY_NAME
-----
PRODUCT_NAME
-----
BRAND
-----
PRICE      STOCK
-----
Guitars
Yamaha Acoustic Guitar
Yamaha
16999      25

CATEGORY_NAME
-----
PRODUCT_NAME
-----
BRAND
-----
PRICE      STOCK
-----
Guitars
Fender Electric Guitar
Fender
45999      10

CATEGORY_NAME
-----
PRODUCT_NAME
-----
BRAND
-----
PRICE      STOCK
-----
Keyboards
Casio Electronic Keyboard
Casio
12999      15
```

Products in Guitar Category

```
SQL> SELECT
  2     p.Product_Name,
  3     p.Brand,
  4     p.Price,
  5     p.Stock
  6 FROM Product p
  7 JOIN Category c
  8 ON p.Category_ID = c.Category_ID
  9 WHERE c.Category_Name = 'Guitars';
```

PRODUCT_NAME

BRAND

PRICE	STOCK
Yamaha Acoustic Guitar	
16999	25

Fender Electric Guitar	
45999	10

PRODUCT_NAME

BRAND

PRICE	STOCK
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Product Count by Category

```
SQL> SELECT
  2     c.Category_Name,
  3     COUNT(p.Product_ID) AS Product_Count
  4 FROM Category c
  5 LEFT JOIN Product p
  6 ON c.Category_ID = p.Category_ID
  7 GROUP BY c.Category_Name;
```

CATEGORY_NAME

PRODUCT_COUNT

Guitars	2
---------	---

Keyboards	1
-----------	---

Microphones	0
-------------	---

CATEGORY_NAME

PRODUCT_COUNT

Amplifiers	0
------------	---

Drums	0
-------	---